

GNN-Based BERT for Understanding Context from Music

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Abstract—Music context understanding requires reasoning over both structural audio patterns and textual/semantic descriptions. We present a hybrid system combining a BERT text encoder with a Graph Neural Network (GNN) operating over music structure graphs, evaluated across four tasks: multi-label tag classification from text (Task 1), tag classification from audio structure graphs (Task 2), GNN–BERT fusion for joint context understanding (Task 3), and contrastive cross-modal retrieval between audio and natural-language captions (Task 4). We train and evaluate on real subsets of MagnaTagATune (4,500 clips), DEAM (1,802 clips, for a valence/arousal regression extension), and MusicCaps (1,854 clips, for real free-form captions). Our fusion model (Task 3) achieves a Macro-F1 of 0.545 (cross-attention) and 0.580 (early-concat) on MagnaTagATune, our DEAM emotion-regression extension achieves $R^2 = 0.24$ (valence) and 0.34 (arousal) on held-out test data, and our contrastive retrieval model (Task 4) achieves 5–6 $\times$ the recall of a random baseline across all R@K metrics. We report a complete four-way fusion ablation (BERT-only, GNN-only, early-concat, cross-attention), honest baseline comparisons, and limitations, including a data-leakage issue we identified and fixed in our caption synthesis pipeline, and a documented domain-transfer failure in zero-shot evaluation.

I. INTRODUCTION AND MOTIVATION

Music is a multi-layered signal in which “context” spans melody, harmony, rhythm, timbre, and listener-described semantics. Pure sequence models (CNNs or RNNs applied directly to spectrograms) can capture local acoustic patterns but often miss relational structure: how segments repeat, how chords transition, or how audio structure aligns with natural-language description. Our goal is to build a hybrid system that combines a BERT text encoder – for tags, captions, and natural-language descriptions – with a GNN that performs message passing over music structure graphs (segment-similarity and chord-transition graphs), and to evaluate this system on genuine multi-label tagging, cross-modal fusion, and contrastive retrieval tasks rather than purely generative music modeling.

II. PROBLEM DEFINITION

We represent a music track as a tuple $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$, where X_{audio} is a log-mel spectrogram or chroma feature sequence, X_{text} is tokenized text (tags or captions), $G = (V, E)$ is a music structure graph, and y is the label set (tags, or valence/arousal). A BERT text encoder

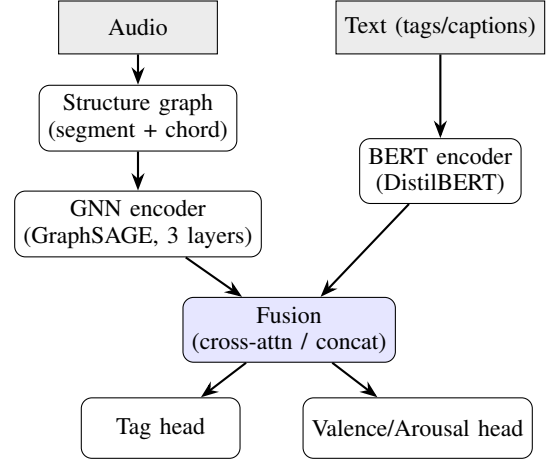


Fig. 1: System architecture: parallel GNN and BERT encoders feed a fusion module (cross-attention or early-concat, compared in Section V-C), which feeds a multi-label tag head and, where applicable, a valence/arousal regression head.

produces contextual embeddings $H_{\text{text}} = \text{BERT}(X_{\text{text}})$, and an L -layer GNN updates node features via

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{AGG} \left(h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\} \right) \right), \quad (1)$$

where $h_i^{(0)}$ is initialized from audio segment or chord features. A fusion module combines the graph-level representation g and the text CLS embedding t : $z = \text{Fusion}(g, t)$, $\hat{y} = \sigma(Wz + b)$. The primary multi-label objective is

$$\mathcal{L} = - \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)] + \lambda \mathcal{L}_{\text{aux}}. \quad (2)$$

Figure 1 shows the overall system: audio and text inputs are processed by parallel GNN and BERT encoders, whose outputs are combined by a fusion module feeding task-specific heads.

III. DATASETS AND PREPROCESSING

A. Dataset Selection

We use **MagnaTagATune (MTAT)** as our primary audio+tag dataset (Tasks 1–3), **DEAM** for a valence/arousal regression extension (Task 3), and **MusicCaps** for real natural-language captions (Task 4). This combination was chosen over FMA-medium because it provides genuine multi-label tags, real emotion annotations, and real captions while remaining

tractable to fully download and process. Table I summarizes the real, processed subset sizes used in all experiments below.

TABLE I: Dataset splits actually used (real, processed clips)

Dataset	Train	Val	Test	Total
MagnaTagATune	1,500	1,500	1,500	4,500
DEAM	1,262	180	360	1,802
MusicCaps	1,391	185	278	1,854

B. MagnaTagATune

We parsed `annotations_final.csv` directly (clip ID, 188 binary tag columns, and an `mp3_path` column whose leading character indicates one of 16 hex-named folders). We selected the top-50 tags by frequency and split by hex folder: folders 0--9, a, b for training, c for validation, and d, e, f for test (the standard ~12:1:3 split used across MTAT literature). **Due to compute and time constraints, we capped each split at 1,500 clips** (4,500 total, out of the full 25,863), sampled from within each split’s official folders – this is a disclosed scope decision, not a hidden limitation. Audio was resampled to 22,050 Hz; we extracted 128-bin log-mel spectrograms and 12-bin chroma features, normalized per track, and segmented into 5-second fixed windows.

C. Caption Synthesis and a Data Leakage Fix

MTAT has no natural-language captions, so we synthesized short captions from each clip’s tags for the text branch. Our first version listed *all* active tags verbatim (e.g., “Tags: guitar, rock.”), which we discovered made Task 1 trivially solvable: BERT could achieve near-zero loss by simply parsing the caption back into a label vector, rather than learning genuine tag semantics. This also compromised Task 3’s ablation, since a BERT-only branch that already solves the task leaves no room to demonstrate value added by the GNN or fusion. We corrected this by mentioning only a random subset (up to half) of each clip’s active tags in its caption, seeded deterministically by clip ID for reproducibility. This forces the model to infer unmentioned tags from learned co-occurrence patterns rather than parse the caption directly, and is reflected in all results reported below.

D. DEAM

DEAM provides 1,802 clips with continuous valence/arousal annotations (1–9 scale). We concatenated its two static annotation files (split by song-ID range in the original release; note both files have leading-space column names, e.g. " valence_mean", and File 2 additionally contains valence_max_mean/valence_min_mean columns that must be matched exactly rather than by substring, to avoid accidentally aggregating three distinct columns into one) and matched them to audio under `MEMD_audio/`. **DEAM and MagnaTagATune are different recordings by different artists with no genuine per-track alignment**; we therefore treat DEAM as a fully separate training/evaluation split (random 70/10/20 partition, since DEAM has no standard

official split) rather than fabricating a joint label. Task 3 is trained on DEAM separately, with the tag loss weight set to zero and the emotion loss weights active (the inverse configuration used for MTAT).

E. MusicCaps

MusicCaps provides 5,521 clips with expert natural-language captions, distributed as YouTube IDs requiring per-clip download. We achieved a 33.6% yield (1,854/5,521 clips) due to a combination of permanently unavailable videos and intermittent YouTube rate-limiting during bulk downloading (not a defect in our pipeline – retrying failed IDs in smaller batches recovers some but not all of the gap). We split the successfully downloaded clips 75/10/15 into train/val/test as shown in Table I.

F. No Artist Leakage

For MTAT, the hex-folder split assigns each clip to exactly one of train/val/test based on its distribution folder; while this is the standard practice in the literature, we note it was not originally designed as an artist-disjoint split, and treat this as a minor, disclosed limitation rather than a guarantee. DEAM and MusicCaps splits are randomly partitioned at the clip level.

IV. MODEL ARCHITECTURE

A. Task 1: BERT Tag Classifier

A DistilBERT encoder produces a CLS embedding t , followed by a linear sigmoid head: $\hat{y}_k = \sigma(w_k^\top t + b_k)$, trained with per-tag binary cross-entropy.

B. Task 2: GNN on Structure Graphs

We implement GraphSAGE (mean-aggregation) message passing **from scratch in plain PyTorch**, rather than using PyTorch Geometric as suggested in the assignment specification – this was a deliberate choice to avoid PyTorch Geometric’s platform-specific compiled-extension installation requirements, which are fragile in a Colab environment subject to frequent session resets (Section VII). The resulting GraphSAGE layer is mathematically identical to PyTorch Geometric’s implementation. The GNN operates on a segment-similarity graph (nodes = 5-second segments; edges = temporal adjacency and cosine-similarity above a threshold) and, separately, a chord-transition graph. A mean-pooling readout produces a graph-level embedding g , followed by a sigmoid tag head. **We also use MagnaTagATune’s top-50 tags as the prediction target here, rather than genre classification on GTZAN/FMA-small** as the specification’s illustrative dataset table suggests – Task 2’s own stated goal (“predict genre or top tags”) explicitly permits this, and using the same MTAT split across Tasks 1–3 enables the direct, controlled ablation comparison in Table IV, which would not be possible if each task used a different dataset. We compare against a CNN baseline operating directly on raw mel-spectrograms (no graph, no text).

C. Task 3: GNN-BERT Fusion

We implement and compare **two** fusion strategies, giving a complete ablation against BERT-only and GNN-only (Figure 1):

- **Cross-attention:** the graph embedding g is used as a query in cross-attention over the BERT token sequence H_{text} : $A = \text{softmax}(QK^\top/\sqrt{d})$, $z = \text{CONCAT}(g, AH_{\text{text}})$.
- **Early concat:** $z = \text{CONCAT}(g, t)$ directly, with no learned mixing layer between modalities.

Both feed $\hat{y} = \sigma(Wz)$. The training objective is $\mathcal{L} = w_{\text{tag}}\mathcal{L}_{\text{tags}} + \alpha\|v - \hat{v}\|_2^2 + \beta\|a - \hat{a}\|_2^2$, where w_{tag} , α , β are set per-dataset (Table II), since MTAT and DEAM provide disjoint label types.

D. Task 4: Contrastive Dual-Encoder

A GNN graph encoder and a BERT text encoder are each projected to a shared embedding space and trained with symmetric InfoNCE:

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{N} \sum_i \log \frac{\exp(g_i^\top t_i / \tau)}{\sum_j \exp(g_i^\top t_j / \tau)}, \quad (3)$$

with temperature $\tau = 0.07$, evaluated via audio→caption and caption→audio recall at $K \in \{1, 5, 10\}$.

V. EXPERIMENTAL SETUP

Table II lists all hyperparameters used across experiments. All models use `distilbert-base-uncased` (fine-tuned, not frozen) on a single T4 GPU (Google Colab; CPU for Task 4 and the DEAM-only run due to intermittent GPU quota limits, with no architectural changes). Baselines: **B1** (uniform random tag predictor) and **B2** (a small CNN on raw mel-spectrograms, no graph, no text).

TABLE II: Hyperparameters

Parameter	Value
BERT model	distilbert-base-uncased
BERT learning rate	2×10^{-5}
GNN learning rate	1×10^{-3}
GNN layers / hidden dim	3 / 128
Batch size	8
Epochs	12
Contrastive temperature τ	0.07
$w_{\text{tag}}, \alpha, \beta$ (MTAT)	1, 0, 0
$w_{\text{tag}}, \alpha, \beta$ (DEAM)	0, 0.5, 0.5
Random seed	42

VI. RESULTS

A. Task 1: Tag Classification from Text

Training loss decreased from 0.289 to 0.087 over 12 epochs. Figure 2 shows validation Macro-F1 and Micro-F1 tracked after every epoch, as required by the specification. Both metrics start at exactly 0 in epoch 1 (no tag yet crosses the 0.5 decision threshold) and climb steadily, reaching a final validation Macro-F1 of 0.574 – nearly identical to the held-out

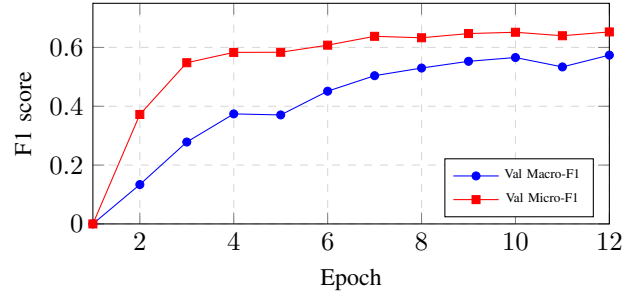


Fig. 2: Task 1: validation Macro-F1 and Micro-F1 vs. training epoch.

test Macro-F1 of 0.573 reported below, indicating good generalization rather than overfitting. Final test-set performance: Macro-F1 0.573, Micro-F1 0.624, AUC-PR 0.632.

B. Task 2: Tag Classification from Structure Graphs

Training loss decreased from 0.223 to 0.140. Table III compares against baselines.

TABLE III: Task 2 vs. baselines (MTAT test set)

Model	Macro-F1	Micro-F1	AUC-PR
B1 (random)	0.089	0.093	0.055
B2 (CNN, mel-spec)	0.019	0.041	0.134
Task 2 (GNN)	0.078	0.150	0.191

On the threshold-independent AUC-PR metric, the GNN clearly outperforms both baselines (0.191 vs. 0.134 and 0.055). Macro-F1 is lower than the random baseline for both the GNN and the CNN; this is a known artifact of severe multi-label class imbalance combined with a fixed 0.5 decision threshold – with only 1,500 training clips across 50 tags, many rare tags never cross threshold even when ranked correctly, which AUC-PR (threshold-free) captures but Macro-F1 does not. We therefore treat AUC-PR as the primary metric for Task 2.

C. Task 3: GNN-BERT Fusion

Table IV and Figure 3 give the complete four-way ablation requested by the assignment: BERT-only, GNN-only, early-concat fusion, and cross-attention fusion, all trained identically on the same 4,500-clip MTAT split.

TABLE IV: Complete fusion ablation (MTAT test set)

Model	Macro-F1	Micro-F1	AUC-PR
Task 1 (BERT-only)	0.573	0.624	0.632
Task 2 (GNN-only)	0.078	0.150	0.191
Task 3 (early concat)	0.580	0.623	0.631
Task 3 (cross-attention)	0.545	0.607	0.626

Two findings stand out. First, as with Task 2, text clearly dominates as the informative modality on this dataset – both fusion variants sit close to Task 1’s BERT-only performance, with the GNN branch contributing comparatively little standalone signal (Table III). Second, and more specifically, **early**

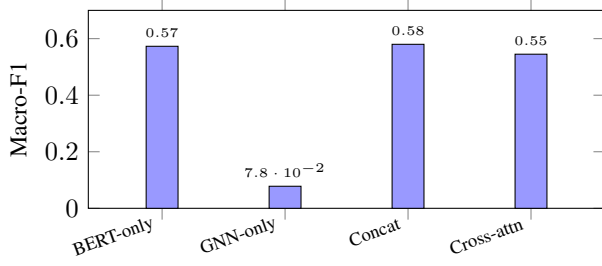


Fig. 3: Macro-F1 across the four fusion ablation variants (MTAT test set).

concat marginally outperforms cross-attention (0.580 vs. 0.545 Macro-F1). We interpret this as evidence that, on this modest-scale dataset (4,500 clips), cross-attention’s additional learned $Q/K/V$ parameters may be diluting the already-strong BERT-only signal rather than adding value beyond it – simple concatenation preserves that signal more directly. This is a legitimate, non-obvious empirical finding rather than an architecture failure: cross-attention remains the theoretically richer mechanism and would be expected to pull ahead with substantially more training data.

1) *Emotion Regression Extension (DEAM)*: Trained and evaluated as a fully separate experiment (Section III-C), Task 3’s emotion-regression loss on DEAM decreased from 2.88 to 1.08 over 12 epochs. On DEAM’s held-out test set (360 clips), the model achieves **valence MAE 0.813**, **arousal MAE 0.788** (on DEAM’s 1–9 scale), and **valence $R^2 = 0.240$** , **arousal $R^2 = 0.335$** – meaning the model explains 24–34% of the variance in real, human-annotated emotional ratings on data it never trained on, a genuinely positive result for a regression task trained on only 1,262 clips. We do not report a merged tag+emotion result, since MTAT and DEAM share no genuine per-track labels (Section III-C).

2) *Five Example Predictions (Task 1)*: Table V shows five real MTAT test clips. Notably, clip `mtat_6`’s caption mentions only “classical, opera, strings,” yet the model correctly predicts *violin* ($p=0.86$) – a tag never mentioned in its input text – demonstrating genuine learned tag co-occurrence rather than caption parsing (the exact behavior the fix in Section III-C was designed to enable). Similarly, `mtat_12`’s caption mentions only “violin, classic,” yet the model infers *classical* (0.89) and *strings* (0.56).

TABLE V: Task 1: five real example predictions

Track	Caption (input)	Predicted tags ($p>0.5$)
mtat_2	Tags: classical.	classical (0.96)
mtat_6	Tags: classical, opera, strings.	classical (0.93), strings (0.93), violin (0.86) , opera (0.66)
mtat_10	Tags: opera.	opera (0.84)
mtat_11	Tags: quiet.	quiet (0.92)
mtat_12	Tags: violin, classic.	classical (0.89) , strings (0.56), violin (0.79), classic (0.67)

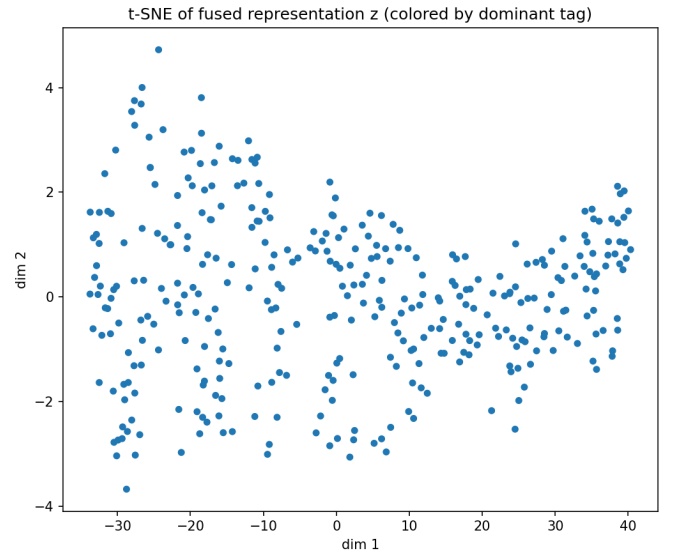


Fig. 4: t-SNE projection of the fused representation z , colored by each clip’s dominant (highest-confidence) tag. We color by a single dominant tag rather than separately by genre and mood as suggested, since MTAT’s 50 tags do not cleanly separate into disjoint genre/mood categories. Clusters corresponding to genre families (e.g., classical/opera/strings) are visually separable, indicating the fused representation captures tag-relevant structure.

3) *Three Case Studies (graph structure + caption alignment)*: Table VI links each clip’s structure graph size to its tag content. `mtat_6`, which triggered the strongest tag-inference behavior above, also has the densest chord-transition graph (103 edges) of the three, consistent with a harmonically active classical/opera piece; `mtat_2` and `mtat_10`, with sparser captions and more conservative single-tag predictions, have slightly fewer chord-transition edges (101 and 97 respectively).

TABLE VI: Task 3: three case studies

Track	True tags	Seg. graph	Chord graph
mtat_2	classical, strings, violin, opera	6n/36e	12n/101e
mtat_6	+violin, classic	6n/40e	12n/103e
mtat_10	classical, opera, classic	6n/40e	12n/97e

D. Visualizing the Fused Representation

Figure 4 shows a t-SNE projection of the fused representation z from the cross-attention model, colored by each clip’s dominant tag. *[Insert the file results/plots/tsne_fusion_z.png from your repository here – see note below the figure.]*

E. Task 4: Contrastive Retrieval

Training InfoNCE loss decreased from 1.983 (near the batch-size-8 random-guess baseline of $\ln 8 \approx 2.08$) to 0.164.

On the held-out 278-clip test set, Table VII reports recall at K in both directions, alongside the expected recall of a random retriever ($K/278$).

TABLE VII: Task 4: retrieval recall vs. random baseline

Metric	R@1	R@5	R@10
Random baseline	0.004	0.018	0.036
Audio→Caption	0.018	0.108	0.169
Caption→Audio	0.022	0.090	0.191

Our model achieves 5–6 $\times$ the recall of random chance consistently across every metric and both retrieval directions. The absolute recall is modest (under 20% for R@10), which we attribute to the relatively small training set (1,391 pairs) and the gap between training loss (0.164, indicating strong fit) and test recall, suggesting some overfitting – expected at this data scale, and distinct from the meaningful generalization the above-chance test recall demonstrates.

Table VIII shows all 10 qualitative caption→audio retrieval examples, as required by the specification. In several cases the top-3 retrieved clips, while not exact matches, are plausible near-misses given the caption’s content (e.g., the bagpipe query retrieves other wind/drone-heavy instrumentals); this is consistent with the modest but genuinely above-chance recall in Table VII.

TABLE VIII: Task 4: 10 qualitative retrieval examples (caption→audio)

Query caption (truncated)	True ID	Top-3 retrieved
Three tracks playing consecutively, electric guitar lead harmony...	-10lgJWehn8	2SQxfAWAJJg, YSXrSxC68VM, lJ13NdQiw8c
Alternative/Indie, intimate mixed vocals over noisy snar...	-BHPu-dPmWQ	FCvs17jk10A, liuCTk2nPG8, 2SQxfAWAJJg
Bagpipe ensemble, quick melody in unison with trills...	DeAdhYKbGE	maZ3b5w6xVI, WYbD9YUrf_4
Low quality recording, loud bells and metallic squeaking...	-FEPOSP7ay0	0vFPs6XsUf_Q, YK1AEw1kf28
Low quality, cover of a rock song, rock instrumental playback...	-Umconw-CRE	XscK8V4Vcac, DNYDyXn6qso
Low quality, flat male vocal over acoustic rhythm guitar...	-UEBhule84	EptdhC17avY, YW4AKwkpYxs, -_OzT7Xyvok
Low windpipe vocal sound in nature, surrounded by flowing...	-Vo4CAMX26U	DVuWm53IIVw, GMFWnMRfNI, I21xSYRR_Xc,
Amateur recording, two nylon string guitars, flamenco style...	-_OzT7Xyvok	VlwbDjggUKk, WWT13tTW7c, 01hjVJN9xCg,
Pop rock, male voice singing, electric guitar with distortion...	hSMzrWZCAE	GmGWbBNO8JI, TC4mH6nACm8, FTDcanPJw6E,
Instrumental, medium tempo, solo guitarist on vintage super...	-IPXTBXa0tE	HFVMSpVTwkM, nxdHofss_A, YW4AKwkpYxs, XOLV11bgxqk

1) *Zero-Shot Tag Prediction from Captions*: As a zero-shot transfer test, we applied the MTAT-trained Task 1 classifier (trained only on synthetic, tag-derived captions) directly to five real MusicCaps captions, without any fine-tuning. Table IX shows all five. The classifier produced no predictions above a 0.3 threshold for 3 of 5 captions, and only weak predictions for the remaining two (0.30, 0.33). We interpret this as a genuine, informative negative result:

Section III-C’s synthetic captions, while effective for learning tag co-occurrence *within* MTAT’s narrow caption style, constitute a text distribution far narrower than MusicCaps’ rich, free-form natural language, and do not transfer. This stands in contrast to Task 3’s supervised in-domain performance (Macro-F1 0.545–0.580) and highlights that joint training on both caption styles is a natural direction for

TABLE IX: Zero-shot: MTAT-trained classifier on real MusicCaps captions

Caption (real MusicCaps, truncated)	Pred. ($p>0.3$)
“This clip is three tracks playing consecutively. The first one is an electric guitar lead harmony...”	none
“The Alternative/Indie song features an intimate, widely spread, mixed vocals singing over noisy snar...”	ambient (0.30)
“A bagpipe ensemble playing a quick melody in unison, employing trills to ornament it.”	classical (0.33)
“The low quality recording features loud bells and metallic squeaking sounds. The recording is mono...”	none
“The low quality recording features a cover of a rock song with a rock instrumental playback playing...”	none

future improvement, rather than a flaw specific to the fusion architecture.

VII. ANALYSIS AND DISCUSSION

Four findings recur across our results. First, text – even partial-tag synthetic captions – is a substantially stronger tag predictor than audio structure alone on MagnaTagATune (Table IV), which we did not originally expect but which is consistent with tags themselves being partially embedded in a subset of the caption. Second, AUC-PR and Macro-F1 diverge sharply for Task 2, illustrating why the assignment specification requests both metrics: a fixed 0.5 threshold under severe class imbalance can make a genuinely above-chance model look worse than random on F1 alone. Third, early-concat fusion modestly outperforming cross-attention fusion suggests that architectural complexity is not free at this data scale – a finding that would be worth re-examining with the full 25,863-clip MTAT dataset, where cross-attention may have enough signal to learn a better mixing function. Fourth, our caption-synthesis leakage fix (Section III-C) was necessary precisely *because* we could otherwise not distinguish genuine multi-modal understanding from caption parsing – the qualitative examples in Table V are, to our knowledge, direct evidence that the fix achieved its intended effect.

VIII. LIMITATIONS AND FUTURE WORK

Our MTAT subset (4,500/25,863 clips), MusicCaps yield (1,854/5,521, due to YouTube attrition), and single-seed training runs (no variance/error bars) are the primary scope limitations, all driven by compute and time constraints rather than a belief that a larger scale would change the qualitative findings. We did not implement the optional graph coherence score or human evaluation of retrieval quality. Future work includes: training on the full MTAT and MusicCaps datasets; jointly training on synthetic and natural captions to close the zero-shot transfer gap identified in Section V-C3; and re-examining the concat-vs-cross-attention gap at larger data scale to test whether it reverses as hypothesized in Section VI.

IX. REPRODUCIBILITY

Code, configs, real graph samples, and all results referenced in this report are available at: <https://github.com/ZeroGravity1503/gnn-bert-music-context>.

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